

Augmentation and leakage

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training a small image model

Without opening the old lesson, explain one key idea from Training a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Keep the equality boundary for Augmentation and leakage. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Augmentation and leakage. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Design a grouped split independently

Six fictional source groups J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to training, M to validation and N, O to final test. Keep all four records per training group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Check the label after a transformation

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

My reasoning and evidence

6. W2 Keep the equality boundary

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

My reasoning and evidence

7. W3 Audit related records

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original. Identify the overlap and explain what must be rerun if this validation set already selected a model.

My reasoning and evidence

8. W4 Design a grouped split independently

Six fictional source groups J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to training, M to validation and N, O to final test. Keep all four records per training group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

My reasoning and evidence

9. W5 Match a training budget

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches. Give both update counts and one way to match budgets without using final outcomes.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled augmentation and leakage evidence sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Augmentation and leakage

Keep every augmented image with its original source group.

Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Augmentation and leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Track provenance

Each flipped copy retains its source ID and parent image reference.



2. Assign by source

A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.



3. Check label meaning

Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

CHANGE ONE THING

Move D's flipped copy into training.

Explain the changed result using the matching lesson.

How many independent source groups are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.